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Original Article

Artificial Intelligence in Sustainable Higher Education: A Systematic Review, Gap Analysis, and Future Research Agenda

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Abstract - *Artificial Intelligence (AI) is increasingly reshaping higher education by transforming teaching and learning processes, assessment practices, institutional governance, and sustainability-related functions. Despite rapid growth in the field, existing research remains fragmented, largely descriptive, and case-based, limiting the development of a coherent and theory-integrated understanding of AI's pedagogical, institutional, and ethical implications within sustainable higher education. This study conducts a systematic literature review following PRISMA guidelines to map the thematic structure and intellectual evolution of AI in sustainable higher education. A structured thematic synthesis of peer-reviewed studies retrieved from major academic databases is employed to examine key application domains, including intelligent tutoring systems, learning analytics, curriculum design, assessment, academic advising, and institutional governance. The review also considers emerging developments in generative AI and their implications for teaching, evaluation, and administrative decision-making. Findings indicate that AI technologies, particularly machine learning, natural language processing, computer vision, and generative AI, enhance automation, personalization, and data-driven decision-making in higher education systems. These capabilities contribute to improved efficiency, scalability, and responsiveness across instructional and administrative processes. However, the analysis also identifies persistent challenges, including algorithmic bias, limited transparency, data privacy risks, and unresolved concerns regarding pedagogical validity and over-reliance on automated systems. The review highlights critical gaps, including limited longitudinal evidence, insufficient cross-context validation, and weak integration between pedagogical theory, technological design, and governance frameworks. Accordingly, a structured future research agenda is proposed, organized around human-centered learning design, assessment validity, ethics and governance, digital equity and infrastructure, AI literacy, theoretical integration, and foresight-oriented research. Overall, the study conceptualizes AI in sustainable higher education as a socio-technical system requiring coordinated advancement in pedagogy, governance, ethics, and infrastructure to ensure sustainable, equitable, and context-sensitive transformation.*

Keywords - *Artificial Intelligence, Sustainable Higher Education, Systematic Review, Learning Analytics, Generative AI, Higher Education Governance.*

1. Introduction

Higher education is undergoing a systemic transformation driven by digital acceleration, knowledge-economy restructuring, and sustainability imperatives, with artificial intelligence (AI) increasingly functioning as a foundational socio-technical infrastructure rather than a peripheral innovation. Across institutional domains—including teaching, learning, research, assessment, and governance AI is embedded in decision-support systems and operational processes, enabling personalization, automation, and data-driven institutional management at scale. This transformation is not only technical but also epistemic and organizational, as it reshapes pedagogical practices, redistributes academic labor, and reconfigures authority structures within universities. While AI enhances efficiency, responsiveness, and innovation capacity, it simultaneously introduces challenges related to epistemic integrity, algorithmic bias, transparency, accountability, data privacy, and equity. Consequently, the adoption of AI requires robust governance frameworks that

align technological advancement with ethical responsibility and universities' human-centered mission (Castillo-Martínez et al., 2024; Deepshikha, 2026; Toha, 2026).

Higher education remains a foundational pillar of societal development, equipping individuals with critical skills and fostering innovation. However, universities worldwide face persistent structural challenges, including inequitable access, limited personalization of learning pathways, and inefficiencies in administrative processes (Fatima et al., 2024; Wasir et al., 2024; Riaz et al., 2026). In response, artificial intelligence (AI) has emerged as a disruptive force capable of improving both educational quality and institutional efficiency (Bearman et al., 2023). These developments are driven by advances in machine learning, natural language processing, predictive analytics, and generative AI, enabling scalable personalization, adaptive learning environments, intelligent automation, and data-driven decision-making (Katsamakas et al., 2024; Almutairi et al., 2025; Afzal et al., 2026).

1.1. AI Applications in Higher Education

Within this evolving landscape, AI is increasingly embedded across core institutional functions, including teaching and learning, student support, assessment, research, and governance. Adaptive learning systems dynamically adjust instructional content to individual student performance, while predictive analytics identify at-risk learners and enable timely interventions (Arshad & Naz, 2024; Naz et al., 2024). Personalized learning is a major application domain where machine learning algorithms tailor content to learners' pace, preferences, and performance (Shafqat et al., 2024). Platforms such as Carnegie Learning and Knewton have demonstrated improvements in engagement and academic achievement through adaptive pathways (Bilal et al., 2025). Intelligent tutoring systems (ITS) further reduce achievement gaps by providing individualized, real-time feedback comparable to one-to-one mentorship (Chatterjee & Bhattacharjee, 2020; Saleem et al., 2024), although their effectiveness depends on institutional infrastructure and faculty training (Zawacki-Richter et al., 2019; Sahar & Munawaroh, 2025).

AI-driven automation is also transforming administrative and assessment processes in higher education. Natural language processing (NLP) tools, including automated essay scoring (AES), reduce faculty workload and enhance grading efficiency (Zafar et al., 2023; Rasheed et al., 2024; Ansari et al., 2025). Systems such as PEG and e-rater demonstrate strong correlations with human grading (Mahar & Zafar, 2025), although they remain limited in assessing creativity and higher-order thinking skills. AI-powered chatbots such as Jill Watson at Georgia Tech handle routine student queries, allowing faculty to focus on more complex academic interactions (Zaffar & Ullah, 2020), although concerns remain regarding the dehumanization of learning environments and potential job displacement.

Despite its benefits, AI integration raises critical ethical, governance, and institutional challenges. Data privacy, algorithmic bias, transparency, and equitable access remain central concerns in the literature (Zawacki-Richter et al., 2019; Ilahi et al., 2023; Arshad, Mehmood & Ahmad, 2025; Bibi et al., 2025; Malik et al., 2025). Regulatory frameworks such as the European Union's General Data Protection Regulation (GDPR) illustrate ongoing tensions between innovation and privacy protection (Bhutto & Zafar, 2023). Algorithmic bias in systems such as facial recognition and AI-based proctoring disproportionately affects marginalized groups (Abbas & Zafar, 2025), reinforcing the need for transparent, inclusive, and accountable AI governance systems (Zawacki-Richter et al., 2019).

Institutional implementation barriers further constrain adoption, including high costs, resistance from educators, and limited AI literacy (Selwyn, 2022; Bonini, 2020). Faculty skepticism regarding AI's pedagogical value highlights the continued importance of human-centered education. In addition, many AI systems function as "black boxes," raising concerns about interpretability, accountability, and trust in decision-making processes (Hassan et al., 2024; Soomro et al., 2024).

Figure 1 conceptualizes AI-driven sustainable higher education as a socio-technical ecosystem in which AI supports teaching and learning, student support, research and innovation, institutional management, assessment, and data-driven decision-making, thereby enhancing adaptability, efficiency, and responsiveness while contributing to sustainability outcomes such as equity, inclusion, resilience, and long-term educational quality. Table 1 presents a staged model of AI integration progressing from digitization (data infrastructure), to automation (process optimization),

personalization (adaptive learning), intelligence (AI-supported decision-making), and transformation (sustainability-oriented institutional redesign).

Despite increasing adoption, existing research remains heavily concentrated on technical dimensions of AI, with limited attention to socio-cultural implications and long-term pedagogical consequences, and is largely dominated by Western contexts with limited representation from the Global South (Lone et al., 2011). These tensions reflect a dual reality in which AI strengthens personalization, efficiency, and responsiveness while simultaneously challenging traditional pedagogical models, particularly regarding human interaction, assessment validity, and academic judgment.

1.2. Research Design, Questions, and Contributions

To address these gaps, this study conducts a PRISMA-guided systematic literature review using Scopus and Web of Science databases. The review maps the intellectual structure and thematic evolution of AI in sustainable higher education and synthesizes applications across intelligent tutoring systems, learning analytics, academic advising, curriculum design, assessment practices, and institutional management.

The study is guided by the following research questions:

- RQ1: How has research on AI in sustainable higher education evolved?
- RQ2: What theoretical frameworks, methodologies, and thematic structures dominate the literature?
- RQ3: How is AI applied across key institutional functions in higher education?
- RQ4: What benefits, limitations, and governance challenges characterize AI adoption?
- RQ5: What critical research gaps remain in the field?
- RQ6: What future research directions are required to advance theory and practice?

This study makes four key contributions. First, it synthesizes fragmented multidisciplinary literature into a coherent analytical framework. Second, it conceptualizes AI as a socio-technical driver of systemic transformation in higher education. Third, it identifies structural gaps in longitudinal, interdisciplinary, governance, and equity-focused research. Fourth, it proposes a structured research agenda emphasizing responsible AI systems, human–AI collaboration, scalable implementation, and sustainability-aligned governance.

The remainder of the paper is structured as follows: Section 2 presents the literature review, Section 3 develops the gap analysis, Section 4 introduces the conceptual framework and research roadmap, and Section 5 concludes the study.

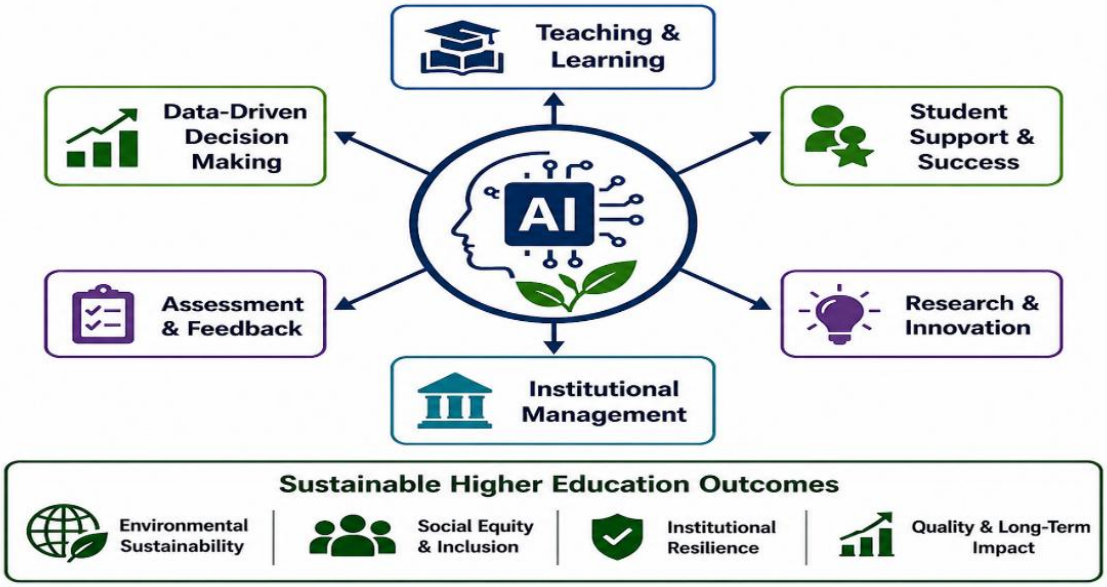


Fig. 1 AI-Driven Sustainable Higher Education

Table 1: AI-Enabled Higher Education Phases.

#	Phase	Objective	Key Activities	AI Capability
1	Digitization	Establish data foundation	Digital records, LMS platforms	Data capture & descriptive analytics
2	Automation	Enhance operational efficiency	Automated grading, chatbots, workflows	Rule-based automation, NLP
3	Personalization	Improve learning experience	Adaptive learning, advising systems	Predictive analytics, recommendation systems
4	Intelligence	Support data-driven decisions	Forecasting, early alerts, planning	Machine learning, decision support
5	Transformation	Achieve sustainable systems	Integrated AI, governance frameworks	AI ecosystems, explainable & responsible AI

2. Literature Review

This study adopts a PRISMA 2020–compliant systematic literature review, complemented by interpretive thematic synthesis, to ensure methodological rigor, transparency, and reproducibility in theory development (Page et al., 2021). A total of over 50 peer-reviewed studies published between 2011 and 2026 were systematically analyzed to examine how artificial intelligence enables sustainable higher education across pedagogical, institutional, and governance dimensions, with particular emphasis on socio-technical transformation, ethical considerations, and long-term sustainability outcomes.

1) Data Sources and Search Strategy: A systematic search was conducted across major academic databases, including Scopus, Web of Science, and ScienceDirect. The search strategy combined artificial intelligence–related terms (e.g., "artificial intelligence," "machine learning," "deep learning," "generative AI") with higher education and sustainability-related terms (e.g., "higher education," "university," "sustainable education," "educational sustainability," "digital transformation"). Boolean operators (AND, OR) were applied to refine and structure the queries, and database-specific filters were used to enhance relevance and precision. The search was limited to peer-reviewed journal articles published in English between 2011 and 2026 to ensure methodological quality and contemporary relevance.
2) Eligibility Criteria: Studies were included if they examined applications of artificial intelligence within higher education contexts and explicitly or implicitly addressed sustainability-related dimensions, including pedagogical innovation, institutional efficiency, equity, governance, or long-term educational impact. Both empirical and conceptual studies were considered when they contributed to understanding AI-enabled transformation in higher education. Studies were excluded if they focused on AI applications outside higher education, lacked relevance to sustainability or systemic transformation, or were non-peer-reviewed sources such as editorials, opinion pieces, or conference abstracts. Additional exclusions were applied to studies with insufficient methodological transparency or limited analytical depth.
3) Screening Process and PRISMA Workflow: The study selection process followed the PRISMA 2020 framework to ensure transparency and reproducibility across the identification, screening, eligibility, and inclusion phases. The initial search yielded 187 records, from which 29 duplicates were removed, leaving 158 records for title and abstract screening. At this stage, 98 records were excluded due to irrelevance, primarily because they did not sufficiently address either artificial intelligence applications or sustainable higher education contexts. The remaining 60 records underwent full-text assessment, after which 7 studies were excluded based on methodological limitations and eligibility criteria. This resulted in a final sample of n = 53 included studies. The complete selection process is presented in the PRISMA flow diagram (Figure 2), ensuring full auditability of the review.
4) Quality Assessment: The methodological quality of the included studies was evaluated using a structured appraisal framework considering clarity of research design, validity and reliability of data sources,

methodological transparency, and relevance to AI-driven sustainable higher education. Particular attention was given to the extent to which studies addressed long-term outcomes, ethical considerations, and sustainability-related impacts. This assessment ensured that only methodologically robust studies informed the synthesis and conceptual development.

5) Data Synthesis: A thematic synthesis approach was employed to systematically identify, analyze, and integrate patterns across the selected studies. The process involved iterative coding, clustering, and theme refinement, enabling both inductive emergence and structured categorization of findings. AI applications were organized into key domains, including intelligent tutoring systems, learning analytics, curriculum design, assessment, academic advising, and institutional governance. Cross-cutting themes such as personalization, efficiency, equity, transparency, and the integration of sustainability were also identified to capture the multidimensional influence of AI in higher education.
6) Methodological Positioning and Conceptual Integration: This study adopts an interpretive and integrative methodological stance that combines systematic review protocols with theory-building objectives. Rather than merely aggregating findings, the analysis develops a socio-technical perspective by linking AI capabilities with pedagogical practices, institutional structures, and governance mechanisms in higher education. This integrative approach supports conceptual advancement by explaining how AI contributes to sustainable transformation in higher education systems and provides the foundation for a structured research roadmap.

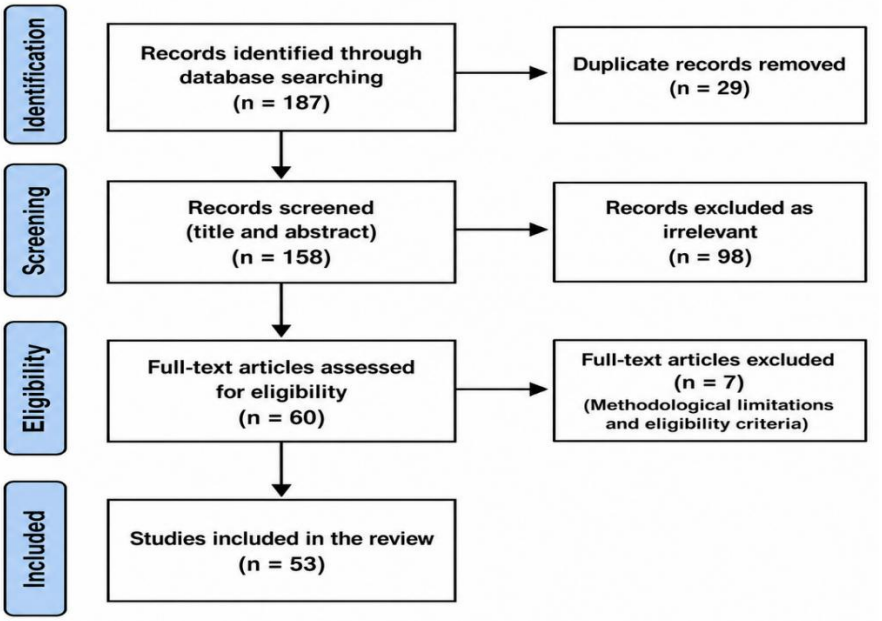


Fig. 2 PRISMA Flow Diagram

2.1. Foundations and Theoretical Evolution of AI in Education

The integration of artificial intelligence into education represents a fundamental paradigm shift toward adaptive, data-driven, and learner-centric systems. This evolution is framed within Artificial Intelligence in Education (AIEd), encompassing intelligent tutoring systems, learning analytics, and human–machine interaction technologies (Chen et al., 2020; Kuleto et al., 2021; Gallent-Torres et al., 2023). Early foundational work by Woolf (1991) introduced intelligent systems designed to model and support human learning processes, later operationalized through Intelligent Tutoring Systems that demonstrated measurable improvements in student motivation and achievement. Subsequently, Woolf (2010) conceptualized AIEd as systems capable of reasoning about learners, content, and pedagogy. Zawacki-Richter et al. (2019) systematized its applications into tutoring, assessment, personalization, and learning analytics.

Recent developments are increasingly shaped by generative AI, which has expanded pedagogical possibilities while simultaneously intensifying concerns regarding academic integrity, accountability, and ethical governance (Crompton & Burke, 2023; Ferdousi, 2024; Hsiao and Tang, 2024). Collectively, these trajectories position AIEd not as a technological enhancement but as a structural reconfiguration of higher education systems.

2.2. AI in Sustainable Higher Education: From Efficiency to Systemic Transformation

Artificial intelligence is increasingly embedded within global sustainability agendas, particularly the United Nations Sustainable Development Goals (SDG 4), which emphasise inclusive and equitable quality education (Chan and Zary, 2019; Rudolph et al., 2024). In higher education, AI is positioned as a transformative enabler for addressing systemic inequalities, resource inefficiencies, and educational accessibility gaps (Toha, 2026).

This has led to the conceptualisation of Sustainable Higher Education (SHE), where AI supports institutional resilience through personalised learning, accessibility enhancement, and resource optimisation (Iatrellis et al., 2024; Tanveer et al., 2020). However, existing literature consistently highlights structural constraints, particularly ethical risks, infrastructural limitations, and governance fragmentation (Alshahrani, 2023; Kooli, 2023). Despite growing adoption, sustainability-oriented AI integration remains largely instrumental rather than systemic.

Theoretically, this study aligns constructivist learning theory with sustainable development theory (Shi et al., 2019), positioning AI as a mediating infrastructure for lifelong learning and sustainability competency development. In this framing, generative AI and learning platforms influence educational sustainability outcomes, while pedagogical design and governance structures moderate these effects. This reflects a shift from technology adoption to socio-technical transformation in higher education systems (Toha, 2026).

2.3. AI-Driven Transformation of Assessment and Personalized Learning

AI has fundamentally restructured assessment systems in higher education by automating evaluation processes, enabling real-time feedback, and expanding learning analytics capabilities (Tapalova and Zhiyenbayeva, 2022). Through natural language processing (NLP) and machine learning (ML), assessment has moved beyond factual recall toward higher-order cognitive skills such as critical thinking, reasoning, and problem-solving (Ramesh and Sanampudi, 2022; Hooda et al., 2022). This aligns with Education 5.0 paradigms that emphasise cybergogy-driven and digitally immersive learning environments.

Empirical evidence indicates that AI enhances grading efficiency, consistency, and perceived fairness while reducing human workload and variability (Schif, 2021). In parallel, adaptive learning systems and AI-based feedback mechanisms improve scalability and instructional responsiveness (Mahmudi et al., 2023; Owan et al., 2023). Large language models further reinforce consistency and perceived objectivity in assessment practices (Ansor et al., 2023). These developments collectively signal a transition from static to dynamic, data-informed assessment ecosystems (Torres, 2023).

Beyond automation, AI enables deep personalization by analyzing learner data to identify cognitive patterns, knowledge gaps, and performance trajectories (Lacey and Smith, 2023). Adaptive systems such as PALAS demonstrate how learning pathways can be dynamically reconfigured based on performance analytics (Palanisamy et al., 2021), while broader intelligent systems align instruction with learner profiles to improve outcomes (Cavanagh et al., 2020; Bates et al., 2020). Learning analytics further strengthen evidence-based decision-making in educational institutions (Sánchez-Prieto et al., 2020; Adiguzel et al., 2023; Hashim et al., 2022). However, these advances remain constrained by unresolved issues related to data privacy, algorithmic bias, transparency, and human oversight (Souppez et al., 2023; Hiran and Singh, 2022; Gamage et al., 2023; Chaudhry and Kazim, 2021; Gardner et al., 2021).

2.4. Core Technologies and AI Applications in Assessment Ecosystems

The technological architecture of AI in education is primarily grounded in natural language processing (NLP), machine learning (ML), and computer vision (CV), which collectively underpin automated assessment systems (Adiguzel et al., 2023). NLP and ML enable structured evaluation of textual outputs through linguistic, semantic, and

conceptual analysis, while simultaneously supporting automated feedback generation. Large language models such as ChatGPT further extend these capabilities by simulating human-like evaluative reasoning (Kooli and Yusuf, 2025; Fernández et al., 2024).

Machine learning enables predictive and adaptive assessment systems through pattern recognition and real-time analytics (Malik et al., 2025; Pang et al., 2024). However, concerns regarding dataset bias and representational inequity remain persistent (Pang et al., 2024). NLP-based automated essay scoring systems improve efficiency and scalability but continue to struggle with contextual sensitivity and pedagogical alignment (Ramesh and Sanampudi, 2022; Memarian and Doleck, 2024). Similarly, computer vision extends AI applications to multimodal assessment tasks, including handwriting and visual analysis, although issues of accuracy, privacy, and computational cost remain significant (Dimari et al., 2024; Lee, 2023; Mahamuni et al., 2024).

Across applications, AI systems include automated grading, automated essay scoring (AES), intelligent tutoring systems (ITS), adaptive learning environments, and feedback generation algorithms (FGA). While these systems improve scalability and efficiency, they remain limited in handling complex reasoning tasks without human intervention (Messer et al., 2024; Kortemeyer et al., 2024). Consequently, hybrid human–AI collaboration is increasingly recognised as essential for ensuring pedagogical validity and fairness (Lyanda et al., 2024; Rudolph et al., 2023).

2.5. Human–AI Balance: Traditional Assessment, Ethics, and Governance

Despite rapid technological adoption, traditional assessment systems remain prevalent due to their interpretive depth, contextual sensitivity, and relational value. However, they are widely criticized for inefficiency, subjectivity, and limited scalability (Ouyang et al., 2023). While AI enhances efficiency and consistency, human assessment provides richer qualitative interpretation and contextual understanding (Ouyang et al., 2023; Dimari et al., 2024; Awidi, 2024; Lee, 2023). Importantly, AI systems may replicate or amplify existing biases embedded in training data, raising concerns about fairness and equity (Stahl et al., 2023; Onesi-Ozigagun et al., 2024; Pang et al., 2024). Despite scalability advantages (Dimari et al., 2024), efficiency-driven assessment risks weakening pedagogical depth (Memarian and Doleck, 2024), while student preference for human evaluation reinforces its continued relevance (Lyanda et al., 2024).

Ethical governance remains a central concern in AI-enabled education. Key challenges include data privacy, transparency, and algorithmic accountability. The extensive use of student data introduces significant privacy risks requiring encryption, secure storage, and regulatory compliance (Onesi-Ozigagun et al., 2024; Lyanda et al., 2024). The inherent opacity of AI systems necessitates explainable AI to ensure trust and accountability (Kooli and Yusuf, 2025). Furthermore, algorithmic bias remains a persistent challenge requiring continuous monitoring and mitigation strategies (Stahl et al., 2023). Effective deployment also depends on robust infrastructure, LMS integration, and institutional readiness (Kortemeyer et al., 2024; Ouyang et al., 2023; Parikh et al., 2023), alongside addressing interoperability barriers (Harry and Sayudin, 2023). AI literacy among educators and students further remains essential for responsible and effective adoption (Ng et al., 2021).

2.6. Research Trends, Bibliometrics, and Future Directions

Bibliometric approaches provide a systematic framework for mapping the intellectual structure, evolution, and impact of AI research in education (Trinh, 2023; Cobo et al., 2011). Performance analysis evaluates scientific productivity through citations and publication metrics, while science mapping examines collaboration networks and thematic structures.

AIEd research has evolved from early instructional systems (Roll and Wylie, 2016) toward automation, personalisation, and ethical governance frameworks (Chen et al., 2020). Contemporary research increasingly focuses on adaptive learning, intelligent tutoring systems, and ethical AI (Guo et al., 2024), while adoption and acceptance studies dominate recent literature (Strzelecki, 2023). Beyond education, AI applications have expanded into healthcare, mental health, and environmental domains (Chen et al., 2024; Ningrum and Muhtar, 2024; Genc and Koçak, 2025), although concerns regarding surveillance, bias, and ethical misuse persist (Borenstein and Howard, 2021).

Future research should prioritize explainable AI, longitudinal evaluation, and comparative assessment of AI versus traditional systems (Wang et al., 2024; Adiguzel et al., 2023). Emerging directions include multimodal assessment, blockchain-based credentials, gamification, and immersive VR learning environments (Zhang, 2024). Advancing these directions requires interdisciplinary collaboration to ensure that AI integration remains ethical, pedagogically grounded, and aligned with sustainability objectives (Stahl et al., 2023; Memarian and Doleck, 2024).

Despite significant advances, critical gaps remain in the literature. First, there is a lack of integrated frameworks connecting AI-driven assessment, personalised learning, and Sustainable Higher Education, despite fragmented but extensive research streams (Tapalova and Zhiyenbayeva, 2022; Lacey and Smith, 2023; Chan and Zary, 2019; Rudolph et al., 2024). Second, comparative empirical evidence across AI technologies and educational contexts remains limited, particularly regarding fairness, bias, and pedagogical validity (Ramesh and Sanampudi, 2022; Pang et al., 2024).

Third, existing research disproportionately prioritises efficiency and automation (Schif, 2021; Mahmudi et al., 2023), while underexploring long-term pedagogical outcomes and the quality of human–AI collaboration (Lyanda et al., 2024; Rudolph et al., 2023). Fourth, although ethical issues are widely acknowledged (Onesi-Ozigagun et al., 2024; Kooli and Yusuf, 2025; Stahl et al., 2023), there remains a lack of empirical validation of governance frameworks and explainable AI implementations in real educational settings. Fifth, AI literacy lacks standardised measurement frameworks for assessing its pedagogical and ethical impact (Ng et al., 2021). Finally, bibliometric studies remain largely descriptive and underdeveloped in predicting future trajectories of AI in sustainable higher education under rapid generative AI expansion (Trinh, 2023; Cobo et al., 2011). These gaps collectively highlight the need for more integrative, longitudinal, and policy-oriented research bridging technology, pedagogy, ethics, and sustainability.

3. Challenges and Research Gaps Analysis

The rapid integration of artificial intelligence into higher education and Sustainable Higher Education (SHE) has progressed at a pace that exceeds the development of coherent pedagogical models, governance structures, and institutional readiness frameworks. As a result, the field is characterised by a productive yet uneven expansion in which technological advancement is outpacing theoretical consolidation, ethical regulation, and long-term educational validation. This imbalance represents a central and persistent tension in Artificial Intelligence in Education (AIEd) and AI-enabled Sustainable Higher Education (Khan et al., 2025).

Figure 2 and Table 2 synthesize the challenges, research gaps, and strategic implications of AI in higher education and SHE by organizing the literature into seven interrelated dimensions. Rather than treating these issues as isolated technological, pedagogical, or institutional problems, it conceptualizes them as a systemic socio-technical ecosystem in which pedagogy, assessment, infrastructure, ethics, human capacity, research integration, and evidence generation are tightly interconnected. This framing highlights that the barriers to AI adoption extend beyond technology to include epistemological, organizational, and governance dimensions, thereby requiring coordinated, system-level responses across higher education ecosystems.

3.1. Pedagogical transformation and learning quality

AI enhances automation, efficiency, and feedback in teaching and assessment systems (Schif, 2021; Mahmudi et al., 2023), yet it also introduces risks of reduced human interaction, weakened formative feedback, and over-reliance on algorithmic mediation. These risks may undermine deep learning processes grounded in constructivist principles, where knowledge is actively constructed through interaction and reflection. The rapid adoption of generative AI further intensifies concerns related to cognitive offloading, reduced intellectual engagement, and academic authenticity (Ferdousi, 2024; Hsiao and Tang, 2024). Despite increasing interest in human–AI collaboration, there remains limited longitudinal evidence on its effects on cognition, learning depth, and pedagogical quality (Lyanda et al., 2024; Rudolph et al., 2023). Strategically, this necessitates redesigning curricula and assessment systems to ensure that critical thinking and epistemic development remain central in AI-enhanced learning environments.

3.2. AI-driven assessment systems and validity concerns

AI-based assessment technologies, including automated essay scoring, intelligent tutoring systems, and large language models, have significantly improved scalability, efficiency, and feedback speed (Ramesh and Sanampudi, 2022; Memarian and Doleck, 2024). However, they still face limitations in contextual understanding, disciplinary sensitivity, and interpretive accuracy, particularly in complex assessment tasks (Pang et al., 2024). In addition, algorithmic bias embedded in training data raises persistent concerns regarding fairness and equity. A key research gap is the lack of robust comparative validation across different AI assessment systems and educational contexts. Strategically, this highlights the need for hybrid assessment models that integrate human judgement with AI support to ensure validity, fairness, and pedagogical alignment.

3.3. Technological infrastructure and system integration

Effective AI implementation depends on robust digital infrastructure, including high-performance computing, scalable storage, and interoperable systems (Kortemeyer et al., 2024; Ouyang et al., 2023). However, many institutions—particularly in developing contexts—face structural limitations in infrastructure readiness, funding, and technical capacity. Moreover, fragmentation between AI tools and Learning Management Systems (LMS) continues to hinder integration and scalability. A major research gap is the absence of infrastructure maturity models that can guide staged and sustainable AI adoption across diverse institutional settings. Strategically, this calls for investment in interoperable and scalable AI ecosystems to reduce global disparities in digital education capacity.

3.4. Ethical, privacy, and governance challenges

AI systems in education rely heavily on student data, raising serious concerns regarding privacy, surveillance, and data misuse (Onesi-Ozigagun et al., 2024; Lyanda et al., 2024). The "black-box" nature of many AI models limits transparency and accountability, making explainable AI a critical requirement for institutional trust and governance (Kooli and Yusuf, 2025). Furthermore, algorithmic bias may reproduce or amplify existing social inequalities (Stahl et al., 2023). Despite widespread recognition of these risks, empirical validation of governance frameworks and explainable AI implementations remains limited. Strategically, this requires the development of institutional AI governance structures, enforceable ethical frameworks, and transparency standards aligned with international regulations.

3.5. AI literacy and human capacity development

Effective AI integration requires educators and students to develop technical, cognitive, ethical, and interpretive competencies (Ng et al., 2021). However, AI literacy levels remain uneven across institutions, disciplines, and regions, leading to inconsistent implementation outcomes. A key research gap is the absence of validated instruments for measuring AI literacy and its influence on pedagogical effectiveness and ethical decision-making. Strategically, this highlights the need to institutionalize AI literacy through curriculum integration and continuous professional development programs.

3.6. Research fragmentation and theoretical integration

Although AI in education has been widely studied, the literature remains fragmented across technological, pedagogical, and sustainability domains (Tapalova and Zhiyenbayeva, 2022; Chan and Zary, 2019; Rudolph et al., 2024). There is still a lack of unified frameworks that integrate AI-driven assessment, personalised learning, and Sustainable Higher Education into a coherent theoretical model. This fragmentation limits cumulative knowledge development and reduces policy transferability. Strategically, this calls for integrated socio-technical frameworks that connect AI, pedagogy, governance, and sustainability in a unified analytical structure.

3.7. Evaluation, evidence generation, and future-oriented research

Current research remains largely efficiency-oriented, focusing on automation, scalability, and performance gains (Schif, 2021; Mahmudi et al., 2023), while underexploring long-term educational outcomes such as learner agency, epistemic development, and the quality of human–AI collaboration (Lyanda et al., 2024). In addition, there is a lack of longitudinal and predictive studies examining how AI transforms higher education over time. Bibliometric analyses, while useful for mapping research trends, remain largely descriptive and insufficient for forecasting future trajectories

under rapid generative AI expansion (Cobo et al., 2011; Trinh, 2023). Strategically, this highlights the need for impact-oriented, longitudinal, and foresight-driven research designs to inform evidence-based policy and institutional transformation.

Overall, these seven dimensions demonstrate that the challenges of AI in higher education are systemic rather than isolated. Addressing them requires a coordinated shift from fragmented technological adoption toward integrated socio-technical systems that align pedagogy, ethics, infrastructure, governance, and sustainability within a coherent and actionable research and policy agenda.

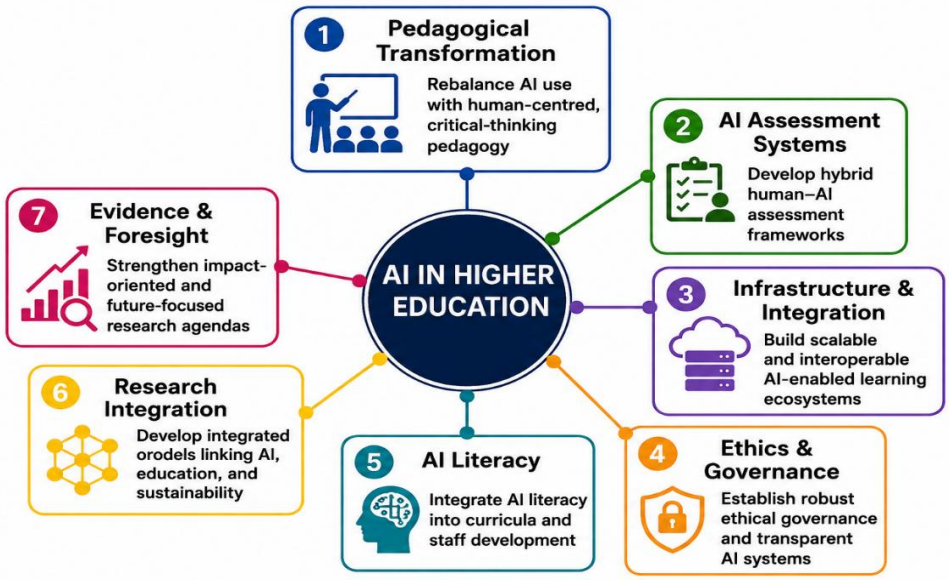


Fig. 2 Key Dimensions and Challenges of AI in Higher Education

Table 2: Challenges, Research Gaps, and Strategic Implications of AI in Higher Education

#	Dimension	Key Challenges	Research Gaps	Strategic Implications
1	Pedagogical transformation	Over-reliance on AI, reduced human interaction, weakened deep learning and feedback quality.	Limited longitudinal evidence on learning impact and cognition	Rebalance AI use with human-centred, critical-thinking pedagogy
2	AI assessment systems	Bias, limited contextual understanding, and fairness/validity concerns.	Insufficient comparative validation across AI assessment tools	Develop hybrid human–AI assessment frameworks
3	Infrastructure & integration	Weak infrastructure, interoperability barriers, and scalability limitations.	Lack of maturity and implementation models for AI integration	Build scalable and interoperable AI-enabled learning ecosystems
4	Ethics & governance	Privacy risks, algorithmic bias, and transparency/accountability gaps.	Limited real-world validation of explainable AI and governance models	Establish robust ethical governance and transparent AI systems
5	AI literacy	Uneven competencies and limited ethical and critical AI awareness.	Absence of standardised AI literacy measurement frameworks	Integrate AI literacy into curricula and staff development

6	Research integration	Fragmentation across pedagogy, technology, and sustainability domains.	Lack of unified socio-technical theoretical frameworks	Develop integrated models linking AI, education, and sustainability
7	Evidence & foresight	Dominance of efficiency-focused studies, limited long-term evaluation.	Lack of longitudinal, predictive, and foresight-oriented research	Strengthen impact-oriented and future-focused research agendas

4. Future Research Agenda

Building on the identified challenges and gaps, the future research agenda for Artificial Intelligence in Higher Education and Sustainable Higher Education (SHE) must shift from fragmented, application-driven work toward an integrated, theory-informed, and system-oriented research paradigm. The current literature shows a persistent imbalance between rapid technological adoption and slower development of pedagogical theory, ethical governance, and institutional readiness. This creates a structural need for a coordinated agenda that aligns AI innovation with educational quality, equity, and sustainability.

Figure 3 and Table 3 consolidate this agenda into seven interrelated research priorities that collectively define the next phase of inquiry. These priorities function as an interconnected socio-technical ecosystem rather than isolated themes, spanning cognition and learning, assessment validity, ethics and governance, infrastructure and equity, AI literacy, theoretical integration, and foresight. Together, they reflect a conceptual shift from studying AI as a set of tools to understanding it as a systemic driver reshaping the structures and practices of higher education.

4.1. Human-centred learning and cognition in AI environments

Future research should prioritise longitudinal evidence on how AI and generative AI reshape cognition, deep learning, learner agency, and knowledge construction. While current studies emphasise efficiency and performance gains, the long-term cognitive and pedagogical implications remain underexplored. Generative AI particularly raises concerns about cognitive offloading, reduced intellectual effort, and academic authenticity, requiring deeper empirical investigation across contexts and disciplines (Schif, 2021; Mahmudi et al., 2023; Ferdousi, 2024; Hsiao and Tang, 2024; Lyanda et al., 2024; Rudolph et al., 2023).

4.2. AI assessment validity and fairness

There is a strong need for systematic validation of AI-driven assessment systems, including automated essay scoring, intelligent tutoring systems, and large language models across diverse educational contexts. Although these systems enhance scalability and efficiency, they remain limited in contextual understanding, interpretive depth, and fairness due to algorithmic bias. Future research should therefore prioritise rigorous comparative studies and the development of hybrid human AI assessment models that balance automation with pedagogical judgement, ensuring validity, equity, and reliability (Ramesh and Sanampudi, 2022; Memarian and Doleck, 2024; Pang et al., 2024).

4.3. Ethics, governance, and explainability

Research must move beyond conceptual discussions toward operational governance mechanisms embedded in institutional practice. Key concerns include data privacy, surveillance risks, algorithmic bias, and lack of transparency in AI systems. The continued reliance on opaque "black-box" models reinforces the need for explainable AI, institutional audit systems, and accountability frameworks that ensure trust, compliance, and ethical alignment in higher education environments (Onesi-Ozigagun et al., 2024; Lyanda et al., 2024; Stahl et al., 2023; Kooli and Yusuf, 2025).

4.4. Infrastructure and digital equity

Sustainable AI integration depends on robust infrastructure, interoperability, and scalable digital systems capable of supporting institutional needs. However, disparities in technological readiness across institutions and regions risk deepening a global AI divide in higher education. Future research should therefore focus on infrastructure maturity

models, scalable implementation pathways, and context-sensitive adoption strategies, particularly for resource-constrained settings (Kortemeyer et al., 2024; Ouyang et al., 2023).

4.5. AI literacy and capacity building

AI literacy should be understood as a multidimensional construct encompassing technical competence, cognitive skills, and ethical judgement for responsible AI use. Despite its importance, current frameworks remain underdeveloped and inconsistently applied. Future research should focus on developing validated measurement instruments and evaluating training interventions that enhance educator capability, student engagement, and responsible AI integration in teaching and learning processes (Ng et al., 2021).

4.6. Theoretical integration and socio-technical frameworks:

The field remains fragmented across technological, pedagogical, and sustainability perspectives, limiting cumulative knowledge development and policy translation. Future research should prioritise the construction of unified socio-technical frameworks that integrate AI systems, learning theories, governance structures, and Sustainable Higher Education outcomes into a coherent conceptual model aligned with global sustainability priorities. Such integration is essential for advancing both theoretical coherence and practical applicability (Tapalova and Zhiyenbayeva, 2022; Chan and Zary, 2019; Rudolph et al., 2024).

4.7. Foresight and impact evaluation

Most existing studies remain descriptive and efficiency-oriented, limiting their capacity to anticipate long-term systemic transformations in higher education. While bibliometric approaches offer useful mapping of research trends, they are insufficient for forecasting rapid developments driven by generative AI. Future research should therefore adopt longitudinal, predictive, and scenario-based methodologies to support evidence-informed policymaking, institutional planning, and sustainable transformation of higher education systems (Schif, 2021; Mahmudi et al., 2023; Cobo et al., 2011; Trinh, 2023).

In synthesis, these seven priorities reposition the future research agenda as a coherent, interdisciplinary, and impact-oriented framework. This is essential to ensure that AI contributes not only to efficiency and innovation but also to pedagogical integrity, institutional resilience, ethical accountability, equity, and long-term sustainability in higher education systems.

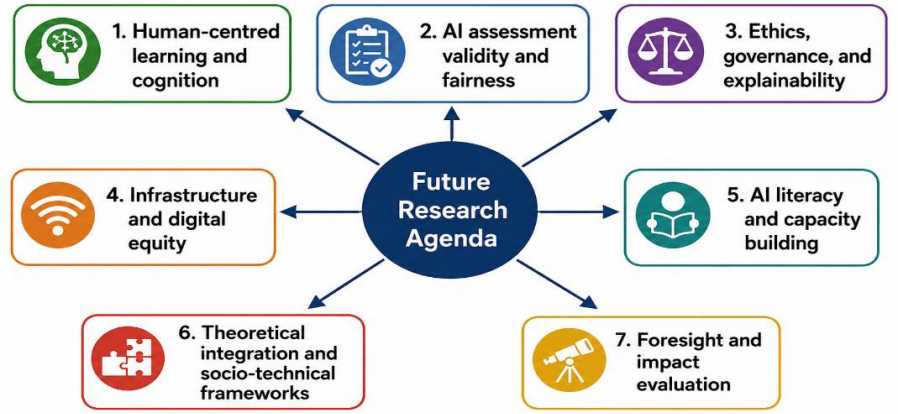


Fig. 3 Future Research Agenda: Seven Interrelated Priorities

Table 3: Future Research Agenda for AI in Sustainable Higher Education.

#	Research Priority	Focus Area	Key Research Directions	Expected Contribution
1	Human-centred learning and cognition	AI-mediated learning processes	Longitudinal studies on cognition, deep learning, learner agency, and generative AI impact	Evidence on how AI reshapes learning depth and academic development

2	AI assessment validity and fairness	Automated and hybrid assessment systems	Cross-context validation of AI assessment; design of hybrid human–AI evaluation models	More valid, fair, and pedagogically aligned assessment systems
3	Ethics, governance, and explainability	Institutional regulation of AI	Operationalisation of explainable AI, audit mechanisms, and accountability frameworks	Transparent, ethical, and trustworthy AI deployment in education
4	Infrastructure and digital equity	AI system readiness and integration	Infrastructure maturity models; LMS interoperability; global access disparities	Scalable and equitable AI-enabled higher education ecosystems
5	AI literacy and capacity building	Competencies of educators and students	Development and validation of AI literacy frameworks; intervention-based training studies	Improved responsible, critical, and effective AI use
6	Theoretical integration	Socio-technical and sustainability frameworks	Unified models linking AI, pedagogy, governance, and sustainability in SHE	Stronger conceptual coherence and policy relevance
7	Foresight and impact evaluation	Long-term transformation of higher education	Longitudinal, predictive, and scenario-based research approaches	Evidence-based forecasting for AI-driven educational transformation

5. Conclusion and Future Work

This study has provided a comprehensive synthesis of Artificial Intelligence (AI) in Higher Education and Sustainable Higher Education (SHE), highlighting its transformative role in reshaping assessment systems, pedagogical practices, and institutional structures. The analysis demonstrates that AI-driven technologies, particularly machine learning, natural language processing, computer vision, and large language models, have significantly enhanced automation, personalization, and learning analytics, leading to improved efficiency, scalability, and responsiveness across higher education systems.

However, the findings also reveal persistent and structurally embedded challenges. These include algorithmic bias, limited transparency, data privacy risks, uneven infrastructure readiness, and concerns regarding pedagogical validity. While AI offers substantial opportunities for improving assessment and learning processes, it cannot fully substitute the contextual sensitivity, ethical judgement, and relational dimensions inherent in human-led education. This reinforces the need for hybrid human–AI approaches that integrate automation with academic integrity and pedagogical quality.

From a sustainability perspective, AI is increasingly positioned as a strategic enabler of inclusive and equitable education systems aligned with global development goals. Nevertheless, its implementation remains uneven due to disparities in institutional capacity, AI literacy, governance maturity, and technological infrastructure. This confirms that AI adoption represents not only a technological shift but also a broader socio-technical transformation requiring coordinated institutional, pedagogical, and policy-level alignment.

Methodologically, the study synthesised existing literature to map key thematic developments and identify critical gaps. The analysis reveals a fragmented research landscape characterised by limited longitudinal evidence, insufficient cross-context validation of AI-based assessment systems, and weak theoretical integration across pedagogical, technological, and sustainability domains. These limitations restrict the development of a cohesive and mature understanding of AI's long-term implications for higher education.

Based on these findings, a structured future research agenda is proposed, organised around seven key priorities: human-centred learning and cognition, AI assessment validity and fairness, ethics and governance, infrastructure and digital equity, AI literacy and capacity building, theoretical integration, and foresight-oriented research. Collectively,

these priorities indicate a necessary shift from isolated technological experimentation toward a coherent, interdisciplinary, and sustainability-aligned research ecosystem.

In conclusion, AI in higher education should be understood not as a standalone innovation, but as a complex socio-technical system that reshapes the foundations of teaching, learning, and institutional governance. Ensuring its responsible and sustainable integration requires sustained interdisciplinary collaboration, strong ethical oversight, and a long-term commitment to educational equity, quality, and institutional resilience.

5.1. Theoretical Implications

This study advances the conceptual understanding of AI in higher education by framing it as a socio-technical system rather than a purely technological enhancement. It integrates constructivist learning perspectives with sustainability-oriented educational theory, highlighting how AI reshapes knowledge construction, learner agency, and institutional transformation. This synthesis helps reduce fragmentation among existing theoretical approaches in AI-enabled education research.

5.2. Practical Implications

The findings provide actionable guidance for educators and instructional designers in developing balanced AI-enhanced learning environments. Emphasis on hybrid human–AI systems supports improved assessment design, more meaningful feedback mechanisms, and responsible integration of AI tools into teaching practices. Institutions can also leverage these insights to strengthen curriculum design and promote AI literacy among students and staff.

5.3. Managerial Implications

For higher education leaders and policymakers, the study underscores the need for structured governance frameworks to support ethical and effective AI adoption. This includes investment in digital infrastructure, development of institutional AI strategies, and establishment of clear accountability and transparency mechanisms. Strategic leadership is essential to ensure that AI implementation aligns with institutional goals, quality assurance standards, and sustainability objectives.

5.4. Study Limitations

This study is based on a structured synthesis of existing literature and does not include primary empirical data collection. Consequently, the findings are constrained by the scope, diversity, and quality of available studies. Additionally, the rapid evolution of AI technologies—particularly generative AI means that some emerging developments may not be fully captured within the analysis.

5.5. Future Research Directions

Future research should prioritise longitudinal and comparative studies to better understand the long-term cognitive, pedagogical, and institutional impacts of AI integration in higher education. There is also a critical need to develop standardized frameworks for AI literacy, governance, and assessment validation, alongside robust models for explainable and accountable AI systems. Furthermore, scenario-based and predictive research approaches should be expanded to anticipate emerging challenges associated with generative AI and its accelerating influence on higher education systems.

Conflicts of Interest:

The authors declare no conflicts of interest.

Data Availability Statement:

All data supporting this study are contained within the article.

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